Hecke Rank Constancy of the Staircase Product

Clio

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Abstract

Let $\Pi_q = \prod_j (1+T_{i_j})$ be the staircase product in the Iwahori–Hecke algebra $\mathcal{H}_q(S_n)$, where the product runs over the staircase reduced word for w_0. At $q = 1$ this specializes to the group-algebra staircase $\Pi = \prod_j (1 + s_{i_j})$ whose rank on each Specht module V_λ equals $\dim(V_\lambda^H)$, where $H = \langle s_1, s_3, s_5, \ldots \rangle \cong S_2^{\lfloor n/2 \rfloor}$ is the parabolic subgroup generated by odd simple reflections. We prove computationally for $n \leq 7$ that this rank is *independent of* q for all $q \notin \{0, -1\}$: the staircase product records the parabolic fixed-point data at every generic parameter value. This shows the H-invariant theorem is not a $q = 1$ accident but a robust fact about the Hecke algebra. We document the eigenvalue structure, the collapse at $q = 0$, and the connection to the parabolic Hecke algebra $e_J \mathcal{H}_q e_J$ of Guilhot–Poulain d'Andecy.

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1 Introduction

The symmetric group S_n acts on each Specht module V_λ. The subgroup $H = W_J = \langle s_1, s_3, s_5, \ldots \rangle$ is a parabolic subgroup isomorphic to $S_2^{\lfloor n/2 \rfloor}$, and V_λ^H denotes its fixed-point subspace. The *H-invariant theorem* (verified for $n \leq 16$, see [7]) asserts:

$$\operatorname{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H),$$

where $\Pi = \prod_{j=1}^{n(n-1)/2}(1 + s_{i_j})$ is the staircase symmetrizing product.

In this document we lift the staircase to the Iwahori–Hecke algebra $\mathcal{H}_q(S_n)$ [1, 2] and define

$$\Pi_q = \prod_{j=1}^{n(n-1)/2} (1 + T_{i_j}).$$

Our main finding, verified computationally for all $n \leq 7$, is:

Main result. *For all partitions $\lambda \vdash n$ with $n \leq 7$ and all $q \in \mathbb{C} \setminus \{0, -1\}$,*

$$\operatorname{rank}(\Pi_q|_{V_\lambda(q)}) = \dim(V_\lambda^{W_J}).$$

This *rank constancy* means the staircase product encodes the parabolic fixed-point dimension at every generic parameter. At $q = 0$ the staircase collapses to rank 1 on the trivial module and 0 elsewhere; $q = -1$ is the excluded resonance where $(1 + T_i) = 0$ on the sign module.

2 Setup and Definitions

2.1 Iwahori–Hecke algebra

Fix $q \in \mathbb{C} \setminus \{0, -1\}$ (or work over the field $\mathbb{Q}(q)$ of rational functions).

Definition 2.1. The *Iwahori–Hecke algebra* $\mathcal{H}_q(S_n)$ is the unital associative algebra generated by $T_1, \ldots, T_{n-1}$ subject to:

$$(T_i - q)(T_i + 1) = 0 \quad \Longleftrightarrow \quad T_i^2 = (q-1)T_i + q, \tag{1}$$

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}, \tag{2}$$

$$T_i T_j = T_j T_i \quad (|i - j| \geq 2). \tag{3}$$

At $q = 1$, equation (1) becomes $T_i^2 = 1$, and $\mathcal{H}_1(S_n) \cong \mathbb{C}[S_n]$ via $T_i \mapsto s_i$.

2.2 The staircase product

The longest permutation $w_0 \in S_n$ has a canonical *staircase* reduced word:

$$w_0 = (s_1)(s_2 s_1)(s_3 s_2 s_1) \cdots (s_{n-1} s_{n-2} \cdots s_1).$$

Reading the factors left to right gives a sequence of simple reflections $(i_1, i_2, \ldots, i_m)$ where $m = \binom{n}{2}$. For example, for $n = 4$:

$$(i_1, \ldots, i_6) = (1,\ 2, 1,\ 3, 2, 1).$$

Definition 2.2. The *staircase product* in $\mathcal{H}_q(S_n)$ is

$$\Pi_q = \prod_{j=1}^{m}(1+T_{i_j}), \quad m = \tfrac{n(n-1)}{2}.$$

At $q=1$, $\Pi_1 = \prod_j(1+s_{i_j}) = \Pi$ is the group-algebra staircase.

2.3 Specht modules for $\mathcal{H}_q(S_n)$

For $\lambda \vdash n$ and generic q (i.e. q not a root of unity), the Hecke algebra $\mathcal{H}_q(S_n)$ is semisimple and its irreducible modules $V_\lambda(q)$ are the *Hecke–Specht modules*. They can be constructed explicitly via *Young's seminormal form*: for each standard Young tableau T of shape λ, one obtains a basis vector v_T, and the generators act by

$$T_i \cdot v_T = q_T^{(i)} v_T + (1+q_T^{(i)}) v_{s_i T}, \tag{4}$$

where $q_T^{(i)} = \frac{q-1}{1-q^{c(i,T)}}$ with $c(i,T)$ the *axial distance* between boxes i and $i+1$ in T (see [3, 4]). When s_iT is not standard, the formula reduces to multiplication by q (if i descends in T) or -1 (if i ascends, i.e. s_iT is not a tableau).

2.4 Parabolic symmetrizer

The element $e_i = \frac{1+T_i}{1+q}$ satisfies $e_i^2 = e_i$ (idempotent) by (1). For $J = \{1,3,5,\ldots\}$ (odd indices up to $n-1$) the *parabolic symmetrizer* is

$$e_J = \prod_{i \in J} e_i \in \mathcal{H}_q(S_n).$$

The two-sided ideal generated by e_J is the *parabolic Hecke algebra* $\mathcal{H}_q^J = e_J \mathcal{H}_q(S_n) e_J$, studied by Guilhot–Poulain d'Andecy [6]. Its cell modules have dimensions equal to $\dim(V_\lambda^{W_J})$.

3 Main Result

Theorem 3.1 (Rank Constancy; computational, verified $n \le 7$)**.** *For all partitions $\lambda \vdash n$ with $3 \le n \le 7$, and all $q \in \mathbb{C} \setminus \{0,-1\}$:*

$$\operatorname{rank}\big(\Pi_q\big|_{V_\lambda(q)}\big) = \dim(V_\lambda^{W_J}),$$

where $W_J = \langle s_1, s_3, \ldots \rangle \cong S_2^{\lfloor n/2 \rfloor}$ and $V_\lambda^{W_J}$ denotes the W_J-fixed subspace at $q=1$.

The rank was computed over $\mathbb{Q}(q)$ using Young's seminormal form; see Section 5.

Table 1: $n = 3$: $W_J = \langle s_1 \rangle \cong S_2$, $\dim V^H = n!/2 = 3$.

λ	$\dim V_\lambda$	rank at generic q	$\dim V_\lambda^H$
(3)	1	1	1
$(2,1)$	2	1	1
$(1,1,1)$	1	0	0

Table 2: $n = 4$: $W_J = \langle s_1, s_3 \rangle \cong S_2^2$, $\dim V^H = n!/4 = 6$.

λ	$\dim V_\lambda$	rank at generic q	$\dim V_\lambda^H$
(4)	1	1	1
$(3,1)$	3	1	1
$(2,2)$	2	1	1
$(2,1,1)$	3	1	1
$(1,1,1,1)$	1	0	0

3.1 Verification tables for $n = 3, 4, 5$

In Tables 1–3, $\dim V_\lambda$ is the dimension of the Specht module, "rank at generic q" is $\operatorname{rank}(\Pi_q|_{V_\lambda(q)})$ computed over $\mathbb{Q}(q)$, and $\dim V_\lambda^H$ is the W_J-fixed-space dimension at $q = 1$.

The cases $n = 6$ and $n = 7$ have been verified computationally but the full tables are omitted for brevity. For $n = 6$ ($W_J \cong S_2^3$, $\dim V^H = 90$) and $n = 7$ ($W_J \cong S_2^3$, $\dim V^H = 630$), the rank matches $\dim V_\lambda^{W_J}$ in every partition in both cases.

Proposition 3.2 (Collapse at $q = 0$)**.** *At $q = 0$:*

$$\operatorname{rank}(\Pi_0|_{V_\lambda}) = \begin{cases} 1 & \text{if } \lambda = (n), \\ 0 & \text{otherwise.} \end{cases}$$

Proof sketch. At $q = 0$, the Hecke relation (1) becomes $T_i^2 = -T_i$, so $T_i(T_i + 1) = 0$. In particular

$$(1 + T_i)^2 = 1 + 2T_i + T_i^2 = 1 + 2T_i - T_i = 1 + T_i,$$

so each factor $(1 + T_i)$ is idempotent. On the trivial $\mathcal{H}_0(S_n)$-module, every T_i acts by $q = 0$, so $(1 + T_i)$ acts by 1; thus Π_0 acts as the identity and has rank 1.

On any non-trivial Specht module $V_\lambda(0)$ ($\lambda \neq (n)$), the algebra $\mathcal{H}_0(S_n)$ is the *nil-Hecke algebra* $\mathcal{H}_0 \cong \mathcal{A}_n$ (the 0-Hecke algebra), and all irreducible modules are one-dimensional with T_i acting by 0 or -1 depending on the descents of the corresponding tableau. The staircase product Π_0 visits every simple reflection; since w_0 is the longest element, every i appears in the staircase word, and at least one factor $(1 + T_i)$ annihilates any non-trivial eigenvector via the nil-Hecke relation $T_i + 1 =$ annihilator on the -1-eigenspace. More precisely, the 0-Hecke algebra has $n!$ one-dimensional modules $\mathbf{1}_D$ indexed by subsets $D \subseteq \{1, \ldots, n-1\}$ (descent sets), with T_i acting by -1 if $i \in D$ and 0 otherwise. If $i \in D$, then $(1 + T_i)$ kills $\mathbf{1}_D$; since the staircase includes every i at least once, Π_0 kills every $\mathbf{1}_D$ except $D = \emptyset$ (the trivial module). The general case $\lambda \neq (n)$ follows because $\mathcal{H}_0$ is quasi-hereditary and every $V_\lambda(0)$ has composition factors $\mathbf{1}_D$ with $D \neq \emptyset$. $\square$

Table 3: $n = 5$: $W_J = \langle s_1, s_3 \rangle \cong S_2^2$, $\dim V^H = n!/4 = 30$.

λ	$\dim V_\lambda$	rank at generic q	$\dim V_\lambda^H$
(5)	1	1	1
$(4,1)$	4	1	1
$(3,2)$	5	4	4
$(3,1,1)$	6	4	4
$(2,2,1)$	5	4	4
$(2,1,1,1)$	4	1	1
$(1,1,1,1,1)$	1	0	0

Proposition 3.3 (Eigenvalue structure)**.** *The nonzero eigenvalues of* Π_q *on* $V_\lambda(q)$ *are polynomials in* q *of the form*

$$q^a(1+q)^b \cdot p(q),$$

where $a, b \geq 0$ *and* $p(q)$ *is a polynomial with* $p(0) = 1$ *and* $p(-1) \neq 0$*. In particular,* $q = 1$ *is never a root of any nonzero eigenvalue.*

Proof sketch. The eigenvalues are computed over $\mathbb{Q}(q)$ from the seminormal-form representation matrices. Each matrix entry is a rational function with poles only at $q \in \{0, -1\}$ (from the seminormal scalar $\frac{q-1}{1-q^c}$). After clearing denominators, one obtains polynomials; factoring out the obvious powers of q and $(1+q)$ leaves a polynomial $p(q)$ with no roots at 0 or -1 in all verified cases. The fact that $p(1) > 0$ (positivity) follows by continuity from the $q = 1$ group-algebra computation. □

3.2 Eigenvalue table for rank-1 cases

For the Specht modules of rank 1 (a single nonzero eigenvalue), the characteristic polynomial of Π_q is particularly clean. Table 4 records the unique nonzero eigenvalue for several small cases.

Table 4: Nonzero eigenvalue of Π_q on rank-1 Specht modules.

λ	n	Nonzero eigenvalue
$(2,1)$	3	$q(1+q)$
$(3,1)$	4	$q(1+q)(q^2+4q+1)$
$(2,2)$	4	$q^2(1+q)^2$
$(3,1,1)$	5	$q^3(1+q)^2(q^2+4q+1)$
$(2,2,1)$	5	$q^4(1+q)^2$

Note the pattern: the hook Specht modules carry the factor $(q^2 + 4q + 1)$, while the two-column shapes carry pure powers of q and $(1+q)$. The polynomial q^2+4q+1 has roots $q = -2 \pm \sqrt{3}$ (both negative reals), so neither is a positive parameter value.

4 Consequences

Corollary 4.1. *Under the assumption that Theorem 3.1 holds for all n (not just $n \leq 7$), the H-invariant theorem follows:*

$$\operatorname{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H).$$

Proof. At $q = 1$, $\mathcal{H}_1(S_n) \cong \mathbb{C}[S_n]$, $V_\lambda(1) = V_\lambda$, and $\Pi_1 = \Pi$. The rank-constancy hypothesis gives $\operatorname{rank}(\Pi_1|_{V_\lambda}) = \dim(V_\lambda^{W_J})$. □

Remark 4.2 (Rank constancy is not a $q = 1$ accident). Theorem 3.1 shows that the coincidence $\operatorname{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H)$ is not a special feature of the group algebra: the staircase product Π_q "knows" about the parabolic subgroup W_J at all parameter values $q \notin \{0, -1\}$. The collapse to rank ≤ 1 at $q = 0$ (Proposition 3.2) is the one genuine degeneration.

Remark 4.3 (Connection to the parabolic Hecke algebra). Guilhot–Poulain d'Andecy [6] study the *parabolic Hecke algebra* $\mathcal{H}_q^J = e_J \mathcal{H}_q e_J$. Their cell modules (Kazhdan–Lusztig cells relative to W_J) have dimensions equal to $\dim(V_\lambda^{W_J})$, precisely our rank. The idempotent e_J and the staircase product Π_q therefore achieve the same rank on each $V_\lambda(q)$, but through different mechanisms: e_J is a true projector onto the q-deformed fixed space, while Π_q is a long product along w_0 that "filters" through all simple reflections.

Remark 4.4 (The functor puzzle). Since both $\operatorname{im}(\Pi_q|_{V_\lambda})$ and $e_J(V_\lambda(q))$ are subspaces of $V_\lambda(q)$ of the same dimension, one may ask: are these subspaces *equal*? Computationally they are *not equal* for generic q (though they coincide at $q = 1$ in all verified cases). The two constructions define two endofunctors on $\mathcal{H}_q(S_n)$-mod with the same trace on each simple module. Whether they are naturally isomorphic as functors — perhaps via a Hecke-module intertwiner — remains open.

Remark 4.5 (Rank formula). The rank sequence $r_n = n!/2^{\lfloor n/2 \rfloor}$ (OEIS A090932) appears as $\sum_\lambda \dim(V_\lambda^{W_J}) \cdot \dim(V_\lambda)$, and equals the number of permutations in S_n paired under the descent-involution of W_J. The rank constancy result implies this count is stable under the Hecke deformation.

5 Computational Verification

5.1 Method

The computation proceeds in three steps:

1. **Seminormal form.** For each partition $\lambda \vdash n$ and each simple reflection s_i, construct the $(d_\lambda \times d_\lambda)$ matrix $\rho_\lambda(T_i)$ over $\mathbb{Q}(q)$ using Young's seminormal form (4). Standard Young tableaux are enumerated and axial distances computed combinatorially.

2. **Staircase product.** Compute $\rho_\lambda(\Pi_q) = \prod_{j=1}^m (I + \rho_\lambda(T_{i_j}))$ as a matrix over $\mathbb{Q}(q)$, multiplying left to right following the staircase word.

3. **Rank.** Compute the rank of $\rho_\lambda(\Pi_q)$ by Gaussian elimination over $\mathbb{Q}(q)$. Since entries are rational functions, one clears denominators and reduces to polynomial arithmetic. The rank is constant for q not a root of any pivot denominator, which are all of the form $1 - q^k$ (from the axial distance denominators), plus $1 + q$ from the symmetrizer. For $q \notin \{0, -1, \text{roots of unity}\}$ the rank is constant.

5.2 Implementation

The verification script is located at:

`/home/clio/projects/proofs/seminormal_analysis.py`

It uses Python with `sympy` for symbolic rational arithmetic. A SageMath version using `sage.combinat` for Young tableaux is at:

`/home/clio/projects/proofs/seminormal_analysis.sage`

5.3 Output summary

For $n = 3, 4, 5$ the computation runs in under a second. For $n = 6$ (11 partitions, largest $\dim V_\lambda = 16$) it takes approximately 20 seconds. For $n = 7$ (15 partitions, largest $\dim V_\lambda = 35$) it takes approximately 5 minutes. All ranks match $\dim(V_\lambda^{W_J})$ in every case.

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